

DAY 3 TASK REPORT

BEGINNER-LEVEL RAG Pipeline Workflow:

Steps :

- (1) User Query - Questions asked by the user
- (2) Document Collection- Source of information
- (3) Text Chunking - Splitting into smaller parts
- (4) Embedding Generation - Converts text into vectors
- (5) Vector Database Storage - stores and search embeddings
- (6) Similarity Search (Retrieval) - find relevant chunks
- (7) Context Retrieval - retrieved chunks are sent to LLMs as context
- (8) Answer Generation - The LLM uses both:
 - Retrieved information
 - Language understanding
- (9) Final Response - answer generate

ADVANCE - LEVEL :

Advanced RAG adds preprocessing, smarter retrieval, and post-processing to handle real-world complexity like noisy data, ambiguous queries, and large document sets.

Steps:

1. Smarter chunking strategies-

Type	When to use
Semantic chunking	Groups related sentences by embedding similarity meilisearch
Recursive/structural chunking	Splits by headings, paragraphs, sections youtube
Parent document retriever	Retrieves small chunks but sends parent context to LLM youtube

2. Hybrid search - keyword+vector search

3. Query rewriting & expansion

- Query rewriting = Rewrites unclear questions into clearer search queries.
- Query expansion = The system automatically expands the user's query.

4. Reranking - After retrieval, a re-ranker sorts results by relevance.

5. Metadata filtering - restricts document retrieval based on metadata attributes before search process.

6. Query router - analyzes a user's query and directs it to the most appropriate data source, retrieval method, or model.